



GRANDICE DOMAIN INTELLIGENCE

TECHNICAL WHITE PAPER • VERSION 1.0

Market Research Intelligence

Deterministic evidence analysis with exact metrics, explicit comparability, contradiction-aware outputs, and caller-attributed provenance.

An implementation-oriented architecture for dated market evidence, competitive landscape analysis, bounded privacy controls, and professional review.

ATTRIBUTE ————— COMPARE ————— DISCLOSE

PUBLISHED

September 2026

METHODOLOGY

grandice-market-deterministic-
1.3.0

STATUS

Initial implementation

Evidence organization—not autonomous market judgment.

Grandice provides two constrained market-research workflows. The first analyzes caller-provided dated evidence for exact metric changes, recency, declared source concentration, contradictions, and research gaps. The second compares declared entities only across fully aligned metric dimensions.

The current implementation contains no model, browser, retrieval, forecasting, ranking, or recommendation step. Source text remains inert data. Every displayed evidence reference carries raw-statement and canonical-record digests, and an analysis manifest binds the methodology, scope, source registry, and evidence set.

Sources remain caller attributed and unverified. Coverage measures the shape of submitted evidence; it does not measure source independence, factual truth, research quality, accuracy, or confidence.

CORE THESIS

A trustworthy market-research workflow must disclose what it received, what is comparable, what conflicts, and what is missing before it attempts to summarize what the market means.

Separate evidence processing from source truth and strategic interpretation.

Research teams routinely combine analyst reports, filings, surveys, announcements, internal observations, and structured metrics. Similar labels may hide different currencies, scales, periods, scopes, or accounting bases. Fresh-looking evidence can repeat one underlying source, while unresolved conflicts disappear inside narrative synthesis.

False comparability

Metrics with matching names are compared despite incompatible units, periods, currencies, or bases.

Source inflation

Multiple caller IDs can describe one publisher or syndicated underlying record.

03

Conflict suppression

A selected "latest" value can hide same-period disagreement or directional ambiguity.

04

Advice drift

Research summaries can become rankings, forecasts, or recommendations without an approval boundary.

Grandice addresses the bounded evidence-processing layer. The adopting organization remains responsible for retrieval, source licensing, authenticity, independence, domain interpretation, methodology approval, decisions, and externally distributed conclusions.

02

SCOPE AND RESPONSIBILITY BOUNDARY

Useful structure without claiming market truth.

Implemented

- Typed caller-provided evidence and research scopes
- Exact Decimal interval arithmetic
- Observation and unique-source recency summaries
- Declared source-ID concentration and HHI
- Disagreement, contradiction, reversal, and gap counts
- Strict cross-entity comparability dimensions
- Canonical evidence, source, and manifest digests
- Killable capacity-limited analysis workers
- Privacy Shield and metadata-only domain audit

Outside current authority

- Live browsing, URL retrieval, or source verification
- Currency, unit, accounting, or period conversion
- Forecasts, causal conclusions, and market predictions
- Objective source quality or independence scoring
- Vendor, company, product, or opportunity rankings
- Investment, market, commercial, or strategic advice
- Price targets, expected returns, or buy/sell/hold language

03

DESIGN PRINCIPLES

Make uncertainty inspectable before making evidence convenient.

01

Caller attribution is not verification

Source labels, dates, publishers, and identities remain caller assertions. Digests bind records; they do not prove truth.

02

Comparability is a schema decision

Name, unit, kind, scale, currency, period, fiscal calendar, and accounting basis must align explicitly.

03

Conflicts remain visible

Disputed endpoints are not selected as authoritative and cannot silently drive trend arithmetic.

04

Coverage is not confidence

A deterministic coverage score reports supplied cells, source IDs, and recency—not accuracy or truth probability.

05

Every output path is bounded

Evidence, rows, bindings, excerpts, response bytes, worker slots, and execution time have enforced limits.

06

Research cannot authorize action

Outputs require professional review and cannot rank, recommend, forecast, trade, purchase, or publish.

A constrained path from protected evidence to inspectable market outputs.

The gateway validates authentication, quotas, JSON bounds, and configured privacy patterns before strict schema validation. Validated analysis then runs inside a capacity-limited child process. A deadline terminates the worker instead of leaving expensive work running in a thread.

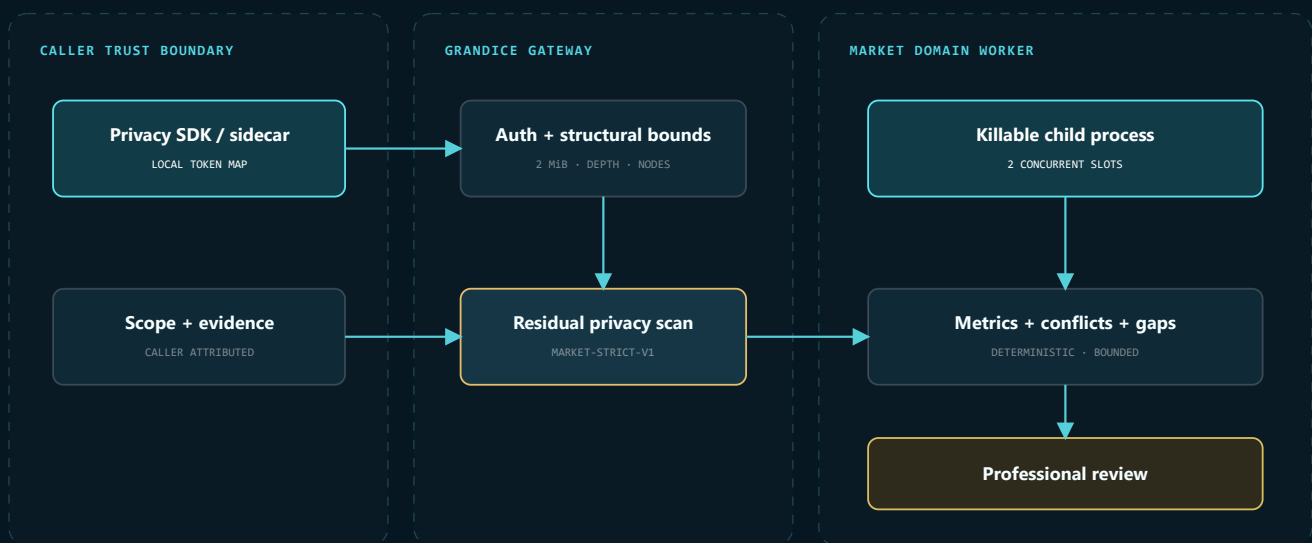


Figure 1. Market evidence never gains control-plane authority. Embedded instructions, HTML, URLs, and recommendation language remain inert input text.

Two deterministic agents with no retrieval or recommendation tools.

EVIDENCE

Market Evidence Analysis

`market.evidence-analyst.v1`

Calculates exact trends, recency, source concentration, coverage, disagreements, contradictions, transitions, and research gaps.

No browsing, source verification, forecasts, ranking, or strategic recommendation.

LANDSCAPE

Competitive Landscape

`market.competitive-landscape.v1`

Builds non-ranking entity profiles and compares only complete, aligned metric dimensions and period boundaries.

No conversion, imputation, "best" vendor, investment attractiveness, or decision.

Registry invariant

`deterministic = true`

`source verification = unverified`

`professional review = required`

Structured observations replace unconstrained research prompts.

06.1 **Research scope**

A scope declares subject IDs, date boundaries, and expected metric dimensions. Metric evidence outside those dimensions is rejected rather than allowed to inflate coverage or suppress declared gaps.

06.2 **Evidence record**

Each item carries an opaque evidence ID, caller source ID, subject, type, observation/publication date, bounded title and statement, evidence kind, signal, tags, and optional complete metric record.

06.3 **Metric semantics**

Metric values declare kind, unit, scale, optional ISO currency, period type, start, end, fiscal calendar, and accounting basis. Floats and incomplete metric tuples are rejected.

06.4 **Date separation**

`observed_date` represents the caller-declared evidence observation/publication date. Metric periods have independent start and end fields. Future-period forecasts are outside this v1 contract.

Exact interval values; explicit quantization where division is required.

Accepted metric numbers contain at most 34 digits, twelve decimal places, and magnitude below 1e25. Interval subtraction uses an 80-digit local Decimal context so accepted values do not inherit ambient 28-digit rounding.

INTERVAL CHANGE $\Delta = V_{\text{end}} - V_{\text{start}}$

Exact for accepted values.

RELATIVE CHANGE $100 \times \Delta / V_{\text{start}}$

Half-even quantized to six places; unavailable at zero or disputed baselines.

JSON FLOATS REJECTED	CAGR NOT INFERRED	DISPUTED ENDPOINTS BLOCKED	PERCENT UNITS EXPLICIT
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A percentage value of "12.5" means 12.5%, not the ratio 0.125. Percentage-point deltas use a distinct value kind and unit.

Describe the submitted evidence without calling it trustworthy.

Observation recency uses caller-declared evidence dates. Source recency uses each referenced source ID once, based on its caller-declared as_of date. Repeating rows from one source cannot improve the recency component.

EVIDENCE COVERAGE SCORE

25% subject cells + 25% expected metric cells + 25% declared source-ID diversity + 25% unique-source recency

Each weighted component contribution is exposed. The result is not a confidence score or truth probability.

Declared source concentration

Counts, largest share, and HHI operate on caller source IDs. Duplicate or syndicated sources can still game this measure.

Research gaps

No evidence, missing dimensions, single-source support, and stale evidence are emitted under reserved quotas.

Completeness

Any row or evidence-binding omission sets complete=false and reports section-level totals.

Disagreement, material conflict, and temporal change are different events.

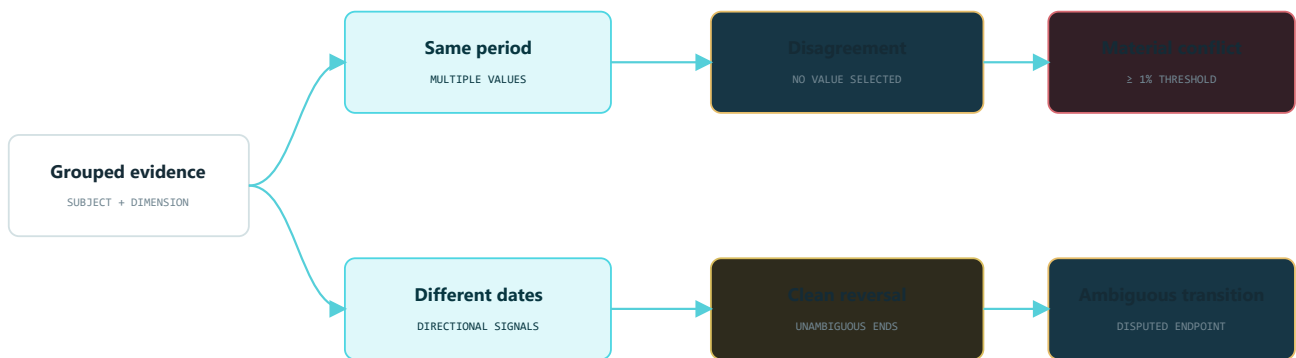


Figure 2. A disputed endpoint prevents authoritative latest-value selection and trend arithmetic. A temporal reversal requires one unambiguous opposing signal at each endpoint.

Matching names are necessary—and nowhere near sufficient.

01	Metric identity name + unit + value kind
02	Numeric basis scale + ISO currency
03	Time basis period type + start + end
04	Reporting basis fiscal calendar + accounting basis
05	Evidence state present + undisputed + known basis

A row becomes comparable only when every required dimension aligns for at least two entities and no entity is missing or disputed. Unknown accounting bases are incomparable. Grandice performs no conversion or imputation.

COMPARABLE	all dimensions aligned	INCOMPLETE	missing entity evidence	INCOMPARABLE	mismatch or dispute
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Bind the analysis record without pretending to authenticate the source.



Figure 3. The manifest detects record substitution across a completed analysis. It does not verify publishers, dates, source independence, licenses, or factual claims.

Protect locally and treat privacy headers as assertions.

Market evidence can contain contacts, account details, confidential strategy, product roadmaps, pricing, acquisition targets, and trade secrets. Local tokenization is the preferred path for confidential workloads. Pattern detection cannot identify every contextual secret.

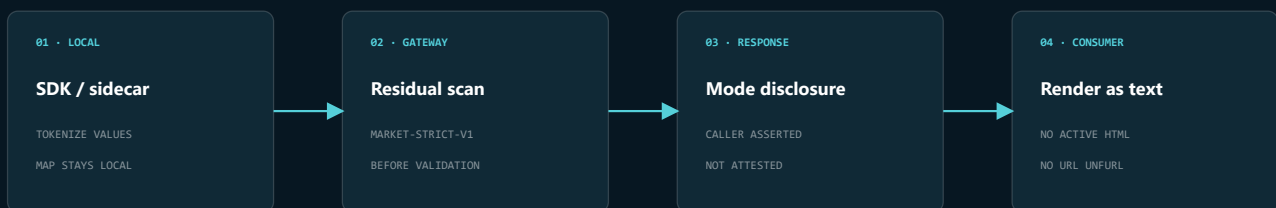


Figure 4. X-Grandice-Privacy-Mode is reported as a caller assertion with `mode_attested=false`. The gateway does not infer that local protection actually occurred.

Resource controls are part of research integrity.

2 MiB REQUEST / RESULT CEILING	2,000 EVIDENCE RECORDS
100 SOURCES / SUBJECTS / DIMENSIONS	25 COMPETITIVE ENTITIES
300 TOTAL SECTION ROWS	8 BINDINGS PER DISPLAYED ROW
160 EXCERPT CHARACTERS	2 WORKER SLOTS PER GATEWAY PROCESS
8s / 10s ANALYSIS / LANDSCAPE DEADLINE	

Latest metrics, trends, contradictions, and gaps receive independent quotas so high-volume metric rows cannot hide critical conflict or gap summaries. Every omission contributes to explicit section counts and sets complete=false.

Operational receipts exclude research content.

Domain audits retain request ID, timestamp, authenticated key and available organization/member references, endpoint, agent, requested privacy mode, status, duration, schema names, source/warning counts, and methodology version. Evidence statements, citations, metrics, excerpts, digests, manifests, and responses are excluded.

domain.runCONTENT RETAINED: FALSE

REQUEST_ID	dom_...
AGENT	market.evidence-analyst.v1
STATUS	200
PRIVACY_MODE	sidecar · asserted
INPUT_SCHEMA	MarketResearchInput
OUTPUT_SCHEMA	MarketResearchAnalysis
METHODOLOGY	grandice-market-deterministic-1.3.0
CONTENT	not stored

Primary domain audit metadata is purged at startup and by hourly opportunistic cleanup after the configured period—90 days by default. Infrastructure logs, crash reporting, and encrypted backups have separate deployment-specific lifecycles.

Direct market APIs or the common managed-agent envelope.

POST	/v1/market/research/analyze
POST	/v1/market/competitive-landscape/compare
GET	/v1/agents
POST	/v1/agents/runs
GET	/v1/domain/runs/{request_id}

```
{
  "metadata": {
    "as_of": "2026-09-15",
    "sources": [{
      "source_id": "filing-a",
      "citation": "Caller repository record",
      "as_of": "2026-09-01"
    }]
  },
  "scope": {
    "scope_id": "software-market",
    "start_date": "2024-01-01",
    "end_date": "2026-09-15",
    "subject_ids": ["entity-a"],
    "metric_dimensions": [{
      "metric_name": "Revenue",
      "metric_unit": "USD",
      "value_kind": "currency",
      "scale": "millions",
      "currency": "USD",
      "period_type": "fiscal_year",
      "accounting_basis": "gaap",
      "fiscal_calendar_id": "dec31"
    }]
  },
  "evidence": ["..."]
}
```

Evaluate arithmetic, coverage, comparability, and reviewer utility separately.

TRACK 01

Data contracts

- Decimal precision and boundary values
- Period-duration and fiscal-calendar validity
- Currency, scale, unit, and basis mismatches
- Canonical digest reproducibility

TRACK 02

Conflict behavior

- Threshold boundary fixtures
- Disputed latest observations
- Clean reversals and ambiguous transitions
- Contradiction quota preservation

TRACK 03

Research coverage

- Declared-dimension cell accounting
- Duplicate source-ID gaming
- Missing entity and one-sided evidence
- Stale and sparse evidence corpora

TRACK 04

Operational safety

- Near-limit request and response bytes
- Worker termination and queue saturation
- Privacy and trade-secret limitations
- Unsafe HTML and URL rendering tests

What this initial evidence engine does not know.

Source limits

- No publisher, author, domain, or license verification
- No retrieval timestamp or independent fetch record
- Source IDs can duplicate or disguise one underlying source
- Citations can be false, stale, incomplete, or biased
- Digests prove record consistency—not factual truth

Analytical limits

- No semantic extraction from unstructured reports
- No causal inference, forecast, or market-size model
- No currency, scope, or accounting conversion
- No source quality or confidence calibration
- No ranking, recommendation, or decision optimization

Privacy limits

- Trade secrets and strategy may evade pattern detection
 - Direct hosted requests expose content to process memory
 - Privacy-mode headers are caller assertions
 - Audit metadata and backups have separate lifecycles
 - Downstream consumers must escape untrusted text
-

CONCLUSION

Reliable market intelligence begins by refusing to hide incompatible or missing evidence.

Grandice combines exact deterministic arithmetic, strict metric comparability, contradiction-aware evidence handling, caller-attributed manifests, bounded process isolation, Privacy Shield integration, metadata-only audit, and mandatory professional review.

The result is an evidence-processing foundation—not autonomous research, verified market truth, or strategic advice. Production adoption still requires approved retrieval, licensing, source validation, representative evaluation, governance, and accountable human interpretation.

EXACT METRICS

EXPLICIT CONFLICTS

HUMAN INTERPRETATION

Response invariants and implementation references

```
{
  "privacy": {
    "mode": "sidecar",
    "mode_attested": false,
    "mode_source": "caller_asserted_header",
    "content_retained": false
  },
  "data": {
    "methodology_version": "grandice-market-deterministic-1.3.0",
    "evidence_provenance": "caller_provided",
    "verification_status": "unverified",
    "coverage": {
      "interpretation": "evidence_coverage_not_truth_probability"
    },
    "complete": true,
    "market_research_advice": false,
    "professional_review_required": true
  }
}
```

<code>grandice_domain/market.py</code>	Evidence schemas, metrics, recency, coverage, conflicts, manifests, and comparison
<code>grandice_domain/agents.py</code>	Versioned market workflow registry and declared capabilities
<code>gateway/routes/market.py</code>	Authenticated APIs, process isolation, deadlines, output ceiling, audit, and headers
<code>gateway/domain_runtime.py</code>	Raw-JSON limits, privacy-mode disclosure, residual checks, and body-free audit
<code>MARKET-APIS.md</code>	Input contracts, limits, privacy guidance, and responsibility boundary
<code>tests/test_market_domain.py</code>	Precision, comparability, conflicts, provenance, bounds, and adversarial tests

IMPORTANT NOTICE

This document describes the September 2026 initial implementation. It is technical product documentation, not investment, market, commercial, strategic, legal, privacy, security, or compliance advice. Sources are caller provided and unverified. Adopting institutions must perform their own validation and professional review.